

Advanced International Macroeconomics B (EI136)

Class 7

Monetary policy: QE/T, new developments

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April 13, 2026

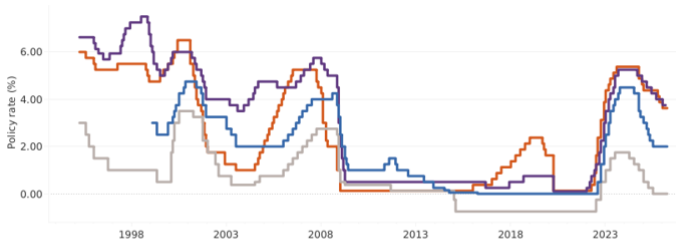
Outline of today class

- Unconventional policy.
 - Overview of impact: theory and evidence (Bhattarai, and Christopher 2022).
 - International spillovers (Georgiadis and Jarocinski 2025).
 - Quantitative tightening (Du, Forbes, Luzzetti 2024).
 - Liquidity dependence (Acharya, Chauhan, Rajan, and Steffen 2022).
- New Keynesian monetary policy models.
 - Trading off of various gaps (Corsetti, Dedola and Leduc 2010).
 - Dealing with FX misalignments from capital flows (Corsetti, Dedola and Leduc 2023).
 - Spillover with financial stability: the FS interest rate (r^{**} , Akinici et al. 2022), and impact on policy rules (Boissay, Collard, Gali, and Manea 2022).

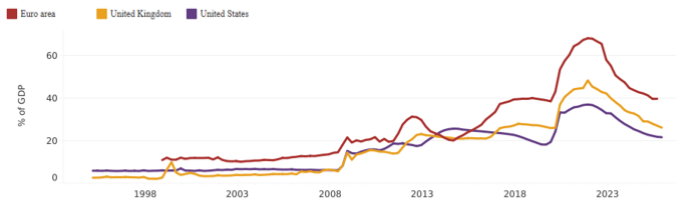
UNCONVENTIONAL POLICY

- Decrease in interest rates and rising balance sheets, offset in recent inflationary episode.

Evolution of the policy rate



Evolution of the total assets



Evolution of the total assets



<https://data.bis.org/topics/CBPOL/tables-and-dashboards>
<https://data.bis.org/topics/CBTA/tables-and-dashboards>

Unconventional Monetary Policy

- Large purchases of assets (Bhattarai, and Christopher 2022).
 - Includes “pure” Quantitative Easing (buy government bonds by creating money, and Credit Easing (change composition of central bank’s assets).
 - Federal Reserve: QE1 in 2009 (buys Treasury and Mortgage Backed Securities), QE2 in 2011 (Treasuries), Maturity Extension in 2012 (move from short- to long-maturity Treasuries), QE3 in 2013-2014 (Treasuries and MBS).
 - ECB more focused on supporting lending to banks, purchases of sovereign bonds since 2016.
- Forward guidance: communicate on future actions.
 - *Delphic* communicates about future economic conditions and *Odyssean* communicates a future path of interest rates.
- Support of lending by banks.
- Negative interest rates.

When can UMP work in theory?

- Why should a central bank buying risky asset matters? Can't private agents do the opposite transaction and arbitrage?
- Segmenting of markets: some households can trade only the short term bond, while the other can trade both bonds, but with at a cost.
 - Without cost, the second households ensures arbitrage.
- Limit to leverage that prevents arbitraging. The Euler relation does not hold any more.
 - Limit can be endogenous (net worth of financial intermediaries).
- Forward guidance affects expectations. Requires history dependence (make up for past mistakes).
- Risk taking channel: improve outlook, raises risk appetite (Bauer et al. 2023).

Empirical evidence: prices

- Event study methodology: look at changes in asset prices at the time of UMP announcements.
- Short windows, but movements in long-term asset prices should reflect the persistent changes in fundamentals.
- BoJ policy supported bank stock prices.
- Effect of forward guidance not clear (hard to isolate).
- More work on effect of large asset purchases.
 - Fed Q1 reduces 10 years yield by 0.55 to 0.91 percent. Bigger effect on mortgage rates.
 - Sizable effect on exchange rate and foreign asset prices (higher bond prices (lower yield), and \$ appreciation).
 - ECB actions reduces 10 years yield by 0.3-0.7 percent.

Empirical evidence: quantities

- QE led to higher volumes of mortgage lending, pushed private portfolios into equity.
- QE2 and 3 led to higher capital flows to emerging economies.
 - Less evidence of capital flows effect of ECB measures.
- ECB measures (TLTRO) helped banks' profits, especially for weaker banks.
- Negative interest rates have not been transmitted so much to loan rates.
- Macroeconomic effects (DSGE, or VAR).
 - DSGE with financial frictions: heterogeneous findings (US GDP increased by 0.13-1 %), sensitive to specific friction, forward guidance appears less effective.
 - ECB TLTRO raised GDP by 1-2.5 %.
- Concerns about limits of UMP.
 - “Wearing off” effect: does impact persist beyond the short event windows? Could reflect a higher growth prospects following UMP.
 - Declining effectiveness as UMP becomes more common. No evidence that effect of later rounds were smaller, controlling for scale.

INTERNATIONAL SPILLOVERS

Spillovers of conventional and unconventional policy

- Assess the impact of Fed policy on the rest of the world financial and real variables (Georgiadis and Jarocinski 2025).
- Contrast conventional interest rate policy, forward guidance, asset purchases (LSAP) and central bank information shocks.
- High frequency identification of asset price movements at the time of FMOOC meetings, and also speeches.
- Identification based on the pattern for different variables (tightening shocks).
 - All raise Treasury yields, Conv. at short maturities, FG and LSAP at longer maturities. Same pattern for expectation component of future short rates.
 - Conv. reduces term premium, LSAP raises it.
 - All reduce stock prices, except CBI with raises it (tighter policy due to a stronger economy).

Pattern of shocks

- Four shocks associated with higher interest rates, but different on maturity and term premium.
- Contrasted effect on stock prices.

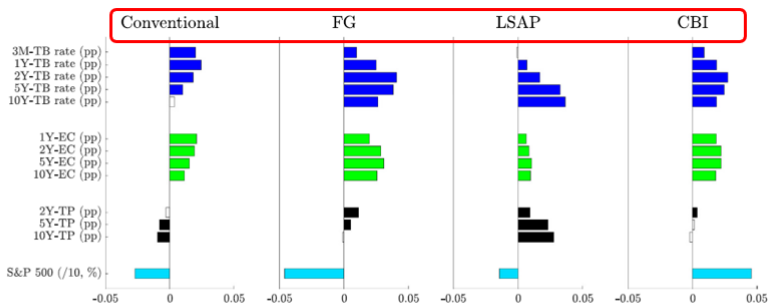


Fig. 1. Impact-day US financial market effects of one-standard-deviation Fed policy shocks.

Georgiadis, Georgios, and Marek Jaroćiński (2025). "Global spillovers from multi-dimensional US monetary policy", *Journal of International Economics* 158.

Impact of the shocks

- Higher interest rates reduce economic activity (except in the CBI case), with strong effect abroad.
- They raise risk aversion and spreads on risky assets.
- Spillover to foreign asset prices, and to capital flows.
 - Large effect on advanced economies bonds. Close substitutes with Treasuries, so higher rates lead US investors from advanced bonds to US ones (preferred habitat).
- Policy trade-off in emerging economies with lower macroeconomic activity and inflation, vs. lower inflows.

Impact on activity

- Reduction of growth (GDP and IP), except for CPI shock.
- Effect similar in the U.S. and abroad.

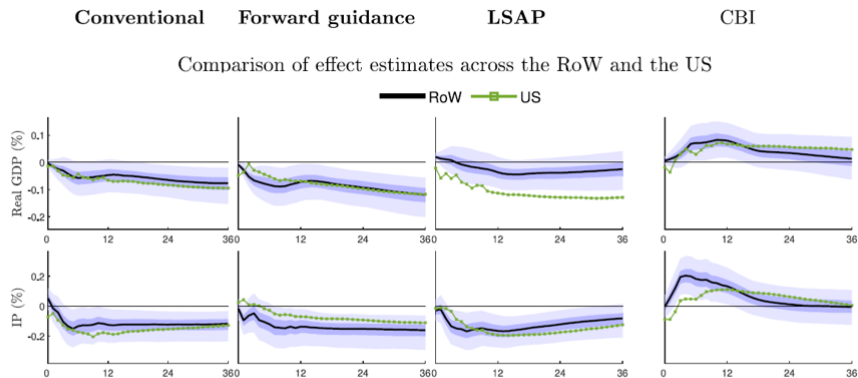


Fig. 3. Effects of US monetary policy shocks on non-US RoW real activity.

Georgiadis, Georgios, and Marek Jarociński (2025). "Global spillovers from multi-dimensional US monetary policy", *Journal of International Economics* 158.

Impact on spreads

- Tighter policy raises the index of **risk aversion**, and raises **high-yield spreads** abroad.

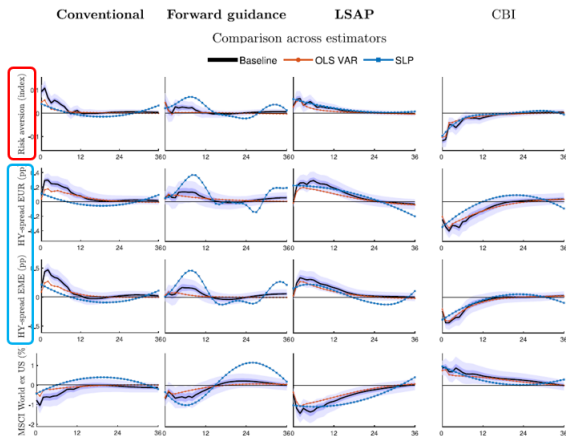


Fig. 5. Effects of US monetary policy shocks on non-US RoW financial variables.

Georgiadis, Georgios, and Marek Jarociński (2025). "Global spillovers from multi-dimensional US monetary policy", *Journal of International Economics* 158.

Impact on capital flows

- Tighter policy reduces US outflows. Contrasted picture with **debt flows** and equity **equity flows**.
- Reduction mostly towards advanced. Conv. reduce all, but LSAP reduces more debt (close substitutes with Treasury bonds).

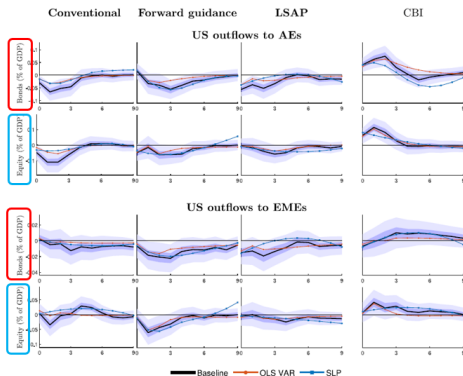


Fig. 7. Effects of US monetary policy shocks on US portfolio outflows by destination and instrument, scaled by lagged US GDP.

Note: See also the note to Fig. 3.

Georgiadis, Georgios, and Marek Jarociński (2025). "Global spillovers from multi-dimensional US monetary policy". *Journal of International Economics* 158.

Policy trade-off

- Tighter policy reduces **growth and inflation** in emerging economies (no trade-off).
- It reduces **inflows** (especially banking). Trade-off between macro stability (lower domestic interest rate) and financial stability (higher rates to maintain inflows).

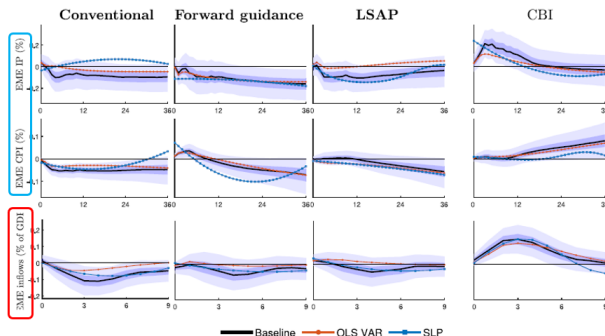


Fig. 15. Effects of US monetary policy shocks on EME variables.

Georgiadis, Georgios, and Marek Jarościński (2025). "Global spillovers from multi-dimensional US monetary policy", *Journal of International Economics* 158.

- Tighter policy leads to a stronger dollar, and lower international trade, including between non-US countries (consistent with prominent dollar role).
- Transmission is stronger for countries that are highly exposed to the U.S. in terms of trade or financial linkages.

QUANTITATIVE TIGHTENING

- Is QT merely a mirror of QE: facts and impact for 7 economies (Du, Forbes, Luzzetti 2024).
 - Large expansions during the Covid pandemic, with subsequent reversion as inflation increased.
- Heterogeneity in implementation and communication of QT.
- Contrast the effect of announcements and actual actions.
 - Database of communication and changes in balance sheets.
 - Active QT or passive (do not roll over maturing securities).
 - Unlike QE, quantitative can be accompanied by interest rate movements.

- Announcements have an effect on sovereign yields, but of limited magnitude (4 bps).
 - Interest rate announcements have more effect.
 - QT effects of specific announcements (68 bps), not general discussion, and larger effect for active QT.
- Limited effect on other financial variables, except for small decrease in convenience yield.
- QT announcements provide signaling of near term interest rate changes (3-6 months).

Impact on financial returns

- QT announcements raise **sovereign yields**, but moderately. Limit effect on **other variables**.

Table 3.1
Announcement Effects of QT on Government Bond Yields

Results for All QT Events (except Wind Downs)						
	Two-day Change in Government Bond Yields (pp)					
	3 month	1 year	2 year	5 year	10 year	30 year
QT Dummy	0.005 (0.010)	0.048** (0.020)	0.037* (0.020)	0.039* (0.020)	0.042** (0.018)	0.040*** (0.014)
Interest Rate Surprise	0.366*** (0.113)	0.445*** (0.122)	0.315*** (0.100)	0.207** (0.081)	0.109 (0.067)	0.002 (0.081)
Economic Data Surprise	0.075 (0.068)	0.492*** (0.059)	0.777*** (0.072)	0.792*** (0.076)	0.603*** (0.074)	0.550*** (0.089)
Observations	9,521	12,534	14,595	14,764	14,858	10,371
R ²	0.006	0.029	0.024	0.017	0.009	0.007

Table 3.2
Announcement Effects of QT on Other Financial Indicators

Results for All QT Events (except Wind Downs)							
	Stock index	Corp Bond index	ER (US\$)	FCI index	Inflation Compensation		Convenience Yield
					3-year	5-year	
QT Dummy	-0.001 (0.002)	-0.002** (0.001)	-0.003 (0.002)	0.023 (0.021)	-0.012 (0.017)	-0.003 (0.013)	-0.020*** (0.007)
Interest Rate Surprise	-0.027*** (0.008)	-0.006* (0.003)	0.020** (0.008)	0.444*** (0.082)	-0.109** (0.055)	-0.034 (0.060)	-0.006 (0.026)
Economic Data Surprise	-0.038*** (0.012)	-0.032*** (0.004)	0.065*** (0.009)	1.076*** (0.088)	0.399*** (0.091)	0.362** (0.172)	-0.067* (0.035)
Observations	14,265	12,201	12,860	14,964	9,694	11,513	14,553
R ²	0.002	0.011	0.008	0.022	0.004	0.003	0.001

Du, Wenxin, Kristin Forbes, and Matthew Luzzetti (2024). "Quantitative Tightening Around the Globe: What Have We Learned?", NBER working paper 32321.

- Essentially no effect of reduction of bond holdings.
 - Letting securities mature should not have an effect, and QT actions are pre-announced.
 - Implementation done with a choice to limit market impact.
- QT on the asset side mirrored by lower bank reserves on the liability side.
 - QT associated with increase in overnight rate spread (4 bps), as liquidity is reduced. Heterogeneous across countries.
 - Reduction of convenience yield on government bonds.
- QT is primarily mirrored by higher holdings of government debt by domestic (and some foreign) non-bank investors. Mirroring is stronger than in non-QT times.

LIQUIDITY DEPENDENCE

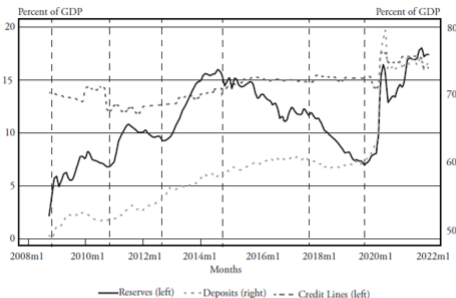
Asymmetric adjustment of banks to QE/QT

- QE (QT) leads to higher (lower) holdings of liquid assets for banks (reserves at the central bank). How have they adjusted (Acharya, Chauhan, Rajan, and Steffen 2022)?
- Increase of short term liquid liabilities.
 - Short term corporate deposits, lines of credit.
 - Reduction in term deposits (shortening of maturity of liabilities).
- Net banks' liquidity has not been changed by QE.
- Adjustment not mirrored in QT: liabilities remained short term.
- Risk of liquidity dependence: QT worsens banks' maturity exposure, may require more QE later on.

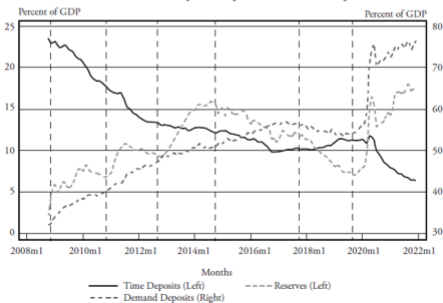
- Deposits and credit lines follow reserves on the way up, but not down.

Chart 1
Time-Series of Aggregate Credit Lines, Deposits and Reserves

Panel A. As percentage of GDP



Panel C. Demand (and other Liquid) Deposits and Time Deposits vs. Reserves



Acharya, Viral, Rahul S Chauhan, [Raghuram Rajan](#), and [Sascha Steffen](#) (2022) "Liquidity Dependence: Why Shrinking Central Bank Balance Sheets is an Uphill Task", Paper presented at the 2022 Jackson Hole symposium.

- Reserves could respond to changing funding structure of banks.
- Use bank-level evidence. Instrument for bank-level funding structure. Some banks are heavier users of reserves.
 - Interact change in overall reserves with *lagged* bank-level reserves, which reflect funding model.
- IV approach shows that during QE higher reserves lead to higher demand deposits and lower time deposits.
- Not mirrored in QT. If anything, demand deposits keep increasing.

- Deposits follow reserves in QE times, but not in QT times (they actually increase further). Time deposits contract in QE, do not revert.

Table 4
Panel B.2
2nd Stage IV Quantity Regressions with $\Delta \text{Ln}(\text{Demand} + \text{Savings Deposits})$
and $\Delta \text{Ln}(\text{Time Deposits})$ as the Dependent Variables

Panel B.2.1	(1)	(2)	(3)	(4)
	$\Delta \text{Ln}(\text{Demand} + \text{Savings Deposits})$	$\Delta \text{Ln}(\text{Demand} + \text{Savings Deposits})$	$\Delta \text{Ln}(\text{Demand} + \text{Savings Deposits})$	$\Delta \text{Ln}(\text{Demand} + \text{Savings Deposits})$
$\Delta \text{Ln}(\text{Reserves})$	0.128*** (0.0168)	0.121*** (0.0172)	0.124*** (0.0147)	-0.134** (0.0677)
$\text{Ln}(\text{Reserves})_{t-5}$	0.0136*** (0.00185)	0.0152*** (0.00246)	0.0156*** (0.00228)	-0.00929** (0.00436)
Obs	115533	50921	43130	30770
Time-FE	Y	Y	Y	Y
Bank & Time Clustered SEs	Y	Y	Y	Bank only
Reg Type	IV	IV	IV	IV
Period	Overall: 2001Q1 - 2021Q4	QE I-III + Pandemic QE: 2008Q4 - 2014Q3 & 2019Q4 - 2021Q4	QE I-III: 2008Q4 - 2014Q3	Post-QE III + QT2014Q4 2019Q3
Panel B.2.2	(1)	(2)	(3)	(4)
	$\Delta \text{Ln}(\text{Time Deposits})$	$\Delta \text{Ln}(\text{Time Deposits})$	$\Delta \text{Ln}(\text{Time Deposits})$	$\Delta \text{Ln}(\text{Time Deposits})$
$\Delta \text{Ln}(\text{Reserves})$	-0.102*** (0.0323)	-0.0831** (0.0309)	-0.0789*** (0.0233)	0.125 (0.175)
$\text{Ln}(\text{Reserves})_{t-5}$	-0.0138*** (0.00388)	-0.0198*** (0.00486)	-0.0175*** (0.00325)	0.0116* (0.0116)
Obs	114689	50555	42853	30551
Time-FE	Y	Y	Y	Y
Bank & Time Clustered SEs	Y	Y	Y	Y
Reg Type	IV	IV	IV	IV
Period	Overall: 2001Q1 - 2021Q4	QE I-III + Pandemic QE: 2008Q4 - 2014Q3 & 2019Q4 - 2021Q4	QE I-III: 2008Q4 - 2014Q3	Post-QE III + QT2014Q4 - 2019Q3

Acharya, Viral, Rahul S Chauhan, Raghuram Rajan, and Sascha Steffen (2022) "Liquidity Dependence: Why Shrinking Central Bank Balance Sheets is an Uphill Task", Paper presented at the 2022 Jackson Hole symposium.

MODELS OF OPTIMAL POLICY

Optimal policy in macro models

- Optimal policy in open economy new Keynesian models (Corsetti, Dedola and Leduc 2010).
- Assess monetary policy rules from an *ex-ante* perspective, with quadratic approximation of utility.
- Optimal stabilization aims at closing various wedges (brings the response at an efficient level).
 - The exchange rate affects the terms-of-trade, but also risk sharing (through the real exchange rate).
 - Closed economy: inflation and output gap (cannot be both zero under cost-push shocks).
 - Open economy: gaps of terms-of-trade, real exchange rate, law of one price, and risk sharing (“demand imbalance”, combines real exchange rate and relative consumptions).
 - Risk sharing gap emerges only if international financial markets are incomplete.
 - Some of the gaps are closed if intertemporal and intratemporal elasticities are equal to 1.

Capital flows misalignments

- How should monetary policy reacts to a surge in inflows that appreciate the currency (Corsetti, Dedola and Leduc 2023)?
- Two country model where capital flows (and other variables) can differ from the first-best allocations. Only one bond is traded (incomplete markets).
 - Capital flows move as agent learn at time t about changes in patience (utility discount factor) from time $t + 1$ onwards.
 - These news do not affect variables in the first best.
- Policy is driven by a serie of gaps (difference from the first best allocation):
 - Real exchange rate gap: $\tilde{Q}_t$ (positive indicates excessive Home real depreciation).
 - Demand gap (cross-country difference of consumption gaps):
 $\tilde{D}_t = \tilde{C}_t - \tilde{C}_t^*$.
 - “Wealth” gap, deviation from efficient risk sharing: $\tilde{W}_t = \sigma \tilde{D}_t - \tilde{Q}_t$.
Positive value indicates Home consumption is too high, or Home currency is too strong ($\tilde{Q}_t < 0$).

Role of exchange rate pass-through

- Optimal policy depends on the impact of exchange rate on import prices.
 - Full pass-through (Producer Currency Pricing): exchange rate movements fully affect import prices.
 - Zero pass-through (Local Currency Pricing): import prices are insulated from the exchange rate.
- Capital inflows shock: Foreign consumers will be more patient in the future.
 - Nothing changes in the first-best before the higher patience actually occurs.
 - Things change under incomplete markets as agents are forward looking: capital flows to Home, real Home appreciation ($\tilde{Q}_t < 0$), $\tilde{W}_t > 0$ irrespective of policy.
- Policy trades off the components of $\tilde{W}_t$.
 - PCP: limit $\tilde{Q}_t$, accept more $\tilde{D}_t$. LCP: opposite pattern.
 - Robust pattern in a more general parametrization, unless trade is very insensitive to relative prices.

- More stabilization of the exchange rate when prices are sensitive to it.

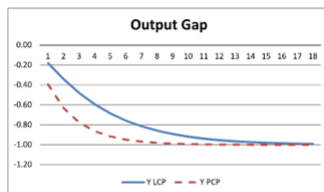
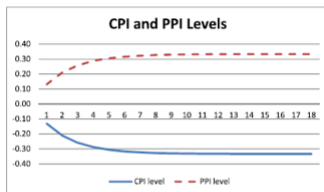
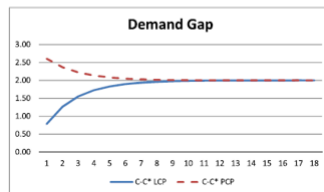
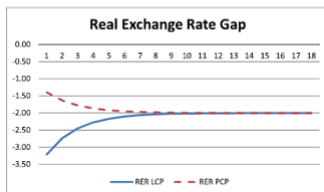
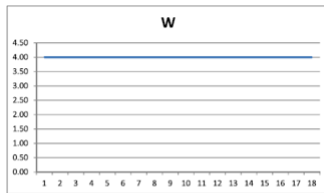
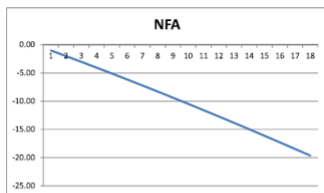


Fig. 1. The figure is drawn for anticipated taste shocks that materialize after period 20 (not shown in the graphs). Parameter values are as follows: $\eta=0$, $\varphi=\sigma=1$, $a_H=.75$, $\beta=.99$, $\alpha=.75$, $\theta=3$.

Corsetti, Giancarlo, Luca Dedola, and Sylvain Leduc (2023). "Exchange rate misalignment and external imbalances: What is the optimal monetary policy response?" *Journal of International Money and Finance* 144.

TAKING ACCOUNT OF FINANCIAL STABILITY

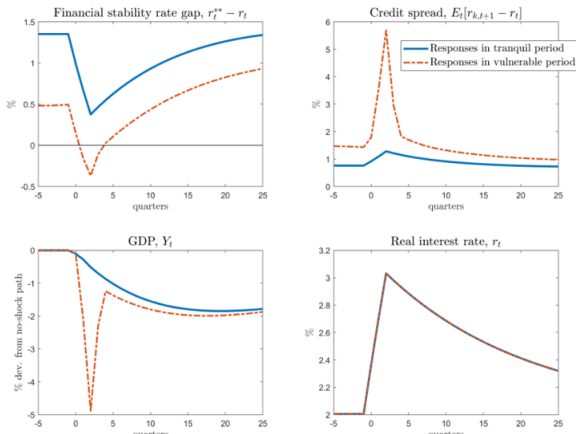
Natural interest rate for financial instability

- Model with occasionally binding credit constraints on banks. What is the interest rate, r_t^{**} , that makes the constraint just binding (Akinci et al. 2022)?
- Sharp recession if an adverse shock pushes r_t^{**} below r_t . This makes the constraint binding.
- Construct an empirical estimates for the US, based on level and volatility of spread on risky returns (distinguish tranquil and crisis times).

Impulse responses

- Higher interest rate shocks leads to a larger recession if r_t^{**} gets below r_t .

Figure 4: Responses to an interest rate shock in financially vulnerable and non vulnerable periods

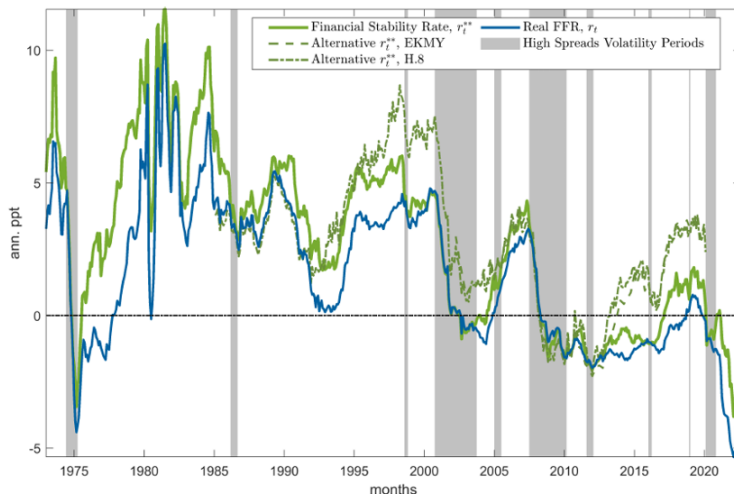


Akinci, Ozge, Gianluca Benigno, Marco Del Negro, and Albert Queralto (2023). "The Financial (In)Stability Real Interest Rate, R^{**} ", Federal Reserve Bank of New York staff report 946.

Estimate of r^{**}

- Downward trend, as for r^* .

Figure 11: The financial stability rate r_t^{**} and the real rate r_t in the data



Akinci, Ozge, Gianluca Benigno, Marco Del Negro, and Albert Queralto (2023). "The Financial (In)Stability Real Interest Rate, R^{**} ", Federal Reserve Bank of New York staff report 946.

SHOULD WE ADJUST TAYLOR RULES?

Model with endogenous crises

- New Keynesian model with heterogeneous firms and credit market frictions (Boissay, Collard, Gali, and Manea 2022).
- Firms buy capital, then learn about idiosyncratic productivity. Efficient allocation: productive firm sell their capital (on credit) to productive firms.
 - Limit on leverage can be binding.
- When credit limit binds, lending market can freeze, leading to recession.
- During booms, capital increases and can be excessive, leading to vulnerability in case of adverse shock.
 - Yield gap: difference between return on equity and its long-run value. A negative yield gap signal inefficient investments (reach for yield) and high crisis risk.

Augmenting Taylor rules

- Should the monetary policy rule be adjusted?
- Stronger reaction to output gap lowers probability of crisis, but makes inflation more volatile and is welfare detrimental.
- Add a reaction the equity yield gap. Lower crisis risk and higher welfare.
 - Not very different during booms, but limits recession depth.
- Backstop rule: standard Taylor rule, but different behavior during crises (large interest rate cut to prevent credit market from freezing), leads to higher welfare.

Comparison of rules

- Some weight on the equity yield gap (augmented Taylor rules) improves welfare.

Table 2: Economic Performance and Welfare Under Alternative Policy Rules

Rule				Model with Financial Frictions					Frictionless
parameters				Time in	Length	Output	Std(π_t)	Welfare	Welfare
ϕ_π ϕ_y ϕ_r				Crisis/Stress (in %)	(quarters)	Loss (in %)	(in pp)	Loss (in %)	Loss (in %)
Taylor-type Rules									
(1)	1.5	0.125	–	[10]	4.8	6.6	1.2	0.82	0.56
(2)	1.5	0.250	–	7.2	4.0	5.4	1.8	1.48	1.21
(3)	1.5	0.375	–	4.1	3.1	4.4	2.5	3.10	2.07
(4)	2.0	0.125	–	9.7	5.0	7.2	0.6	0.41	0.17
(5)	2.5	0.125	–	9.6	5.1	7.5	0.5	0.31	0.08
SIT									
(6)	$+\infty$	–	–	9.4	5.1	8.1	–	0.23	0.00
Augmented Taylor-type Rules									
(7)	1.5	0.125	5.0	5.4	3.9	5.5	1.16	0.65	–
(8)	5.0	0.125	5.0	8.8	5.0	7.4	0.18	0.22	–
(9)	5.0	0.125	25.0	6.9	4.7	6.6	0.19	0.18	–
(10)	10.0	0.125	75.0	6.3	4.6	6.4	0.09	0.16	–
Backstop Rules									
(11)	1.5	0.125	–	15.5	–	–	1.21	0.56	–
(12)	$+\infty$	–	–	17.1	–	–	0.50	0.10	–

Boissay, Frederic, Fabrice Collard, Jordi Gali, and Cristina Manea (2022) "Monetary policy and endogenous financial crises", BIS working paper 991.